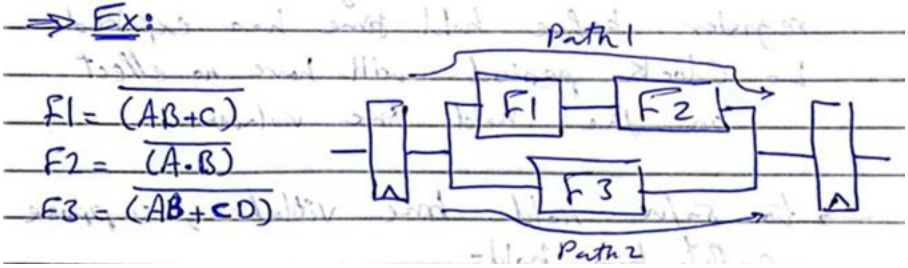
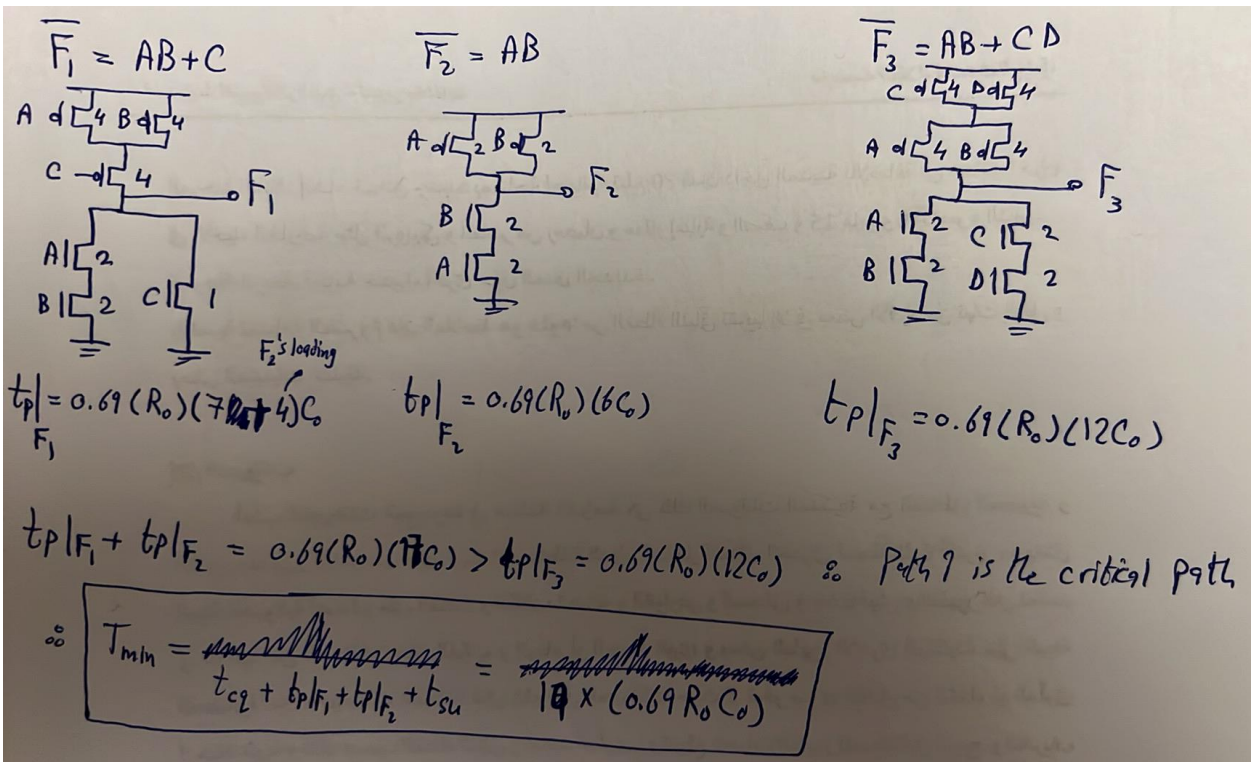


(1) Assuming $t_{cq} = t_{su} = 0.69R_oC_o$, find the minimum clock period (assume registers loading is negligible).

Answer:

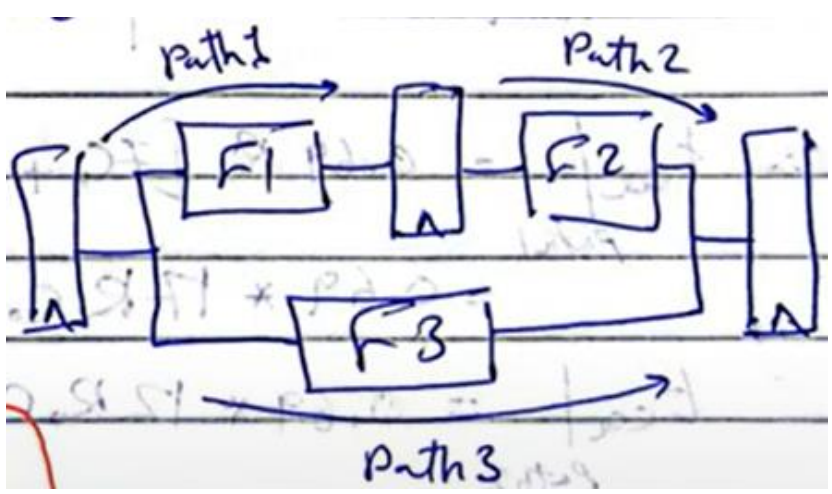
To know the minimum clock period, we need to identify the critical path using the worst-case delays:



(2) Assume that the clock period can't be more than $0.69 * 14R_oC_o$, what type of violation do we have? Slack? How to solve it?

Answer:

T_{min} is $0.69 * (19R_oC_o)$ while the used clock period is $0.69 * 14R_oC_o$. Thus, there is a setup-time violation with a negative slack of $0.69 * 6R_oC_o$. To solve this, break the critical path, which is path 1, as follows:



$$t_p|_{path\ 1} = 0.69 * (R_o)(7C_o)\ (no\ F2's\ loading\ anymore)$$

$$t_p|_{path\ 2} = 0.69 * (R_o)(6C_o)$$

$$t_p|_{path\ 3} = 0.69 * (R_o)(12C_o)$$

Thus, path 3 becomes the critical path with critical path delay of:

$t_{cq} + t_{path\ 3} + t_{su} = 0.69R_oC_o + 0.69(R_o)(12C_o) + 0.69R_oC_o = 0.69 * (14R_oC_o) = T_{min}$ ✓

(3) What if $t_{clk} = 0.69 * 10R_oC_o$?

Answer:

The only solution here is to somehow break path 3 and do internal pipelining *<u>if possible.</u>* If not possible, the design or clock period must be changed.

(4) Assume $t_{hold} = 0.69 * 4.25R_oC_o$ after the situation in part (2), do we have any violations? How to solve them?

Answer:

For hold-analysis, we calculate the best-case delays:

$$t_p|_{F_1,best} = t_{phl}|_{F_1,best} = 0.69(0.5R_o)(7C_o) = 0.69(3.5R_oC_o)$$

$$t_p|_{F_2,best} = t_{plh}|_{F_2,best} = 0.69(0.5R_o)(6C_o) = 0.69(3R_oC_o)$$

$$t_p|_{F_3,best} = t_{phl}|_{F_1,best} = t_{plh}|_{F_1,best} = 0.69(0.5R_o)(7C_o) = 0.69(3.5R_oC_o)$$

$$arrival\ time|_{path1} = t_{cq} + t_p|_{F_1} = 0.69(4.5R_oC_o)$$

$$arrival\ time|_{path2} = t_{cq} + t_p|_{F_2} = 0.69(4R_oC_o)$$

$$arrival\ time|_{path3} = t_{cq} + t_p|_{F_3} = 0.69(4.5R_oC_o)$$

$$Required\ time = t_{hold} = 0.69(7R_oC_o)$$

Since we want $arrival\ time > required\ time \rightarrow$ path 2 has a hold-time violation. How to solve it? By adding two unit inverters having a delay of $0.69 * 3R_oC_o$ in path 2. However, we need to check back that this does not cause a setup violation:

$$\begin{aligned} t_p|_{path_2,worst} &= t_{cq} + \left(t_p|_{F_2,worst} + 2 * t_p|_{inverter}\right) + t_{su} \\ &= 0.69R_oC_o + [0.69 * (R_o)(6C_o) + 2 * 0.69 * 3R_oC_o] + 0.69R_oC_o = 0.69 * (14R_oC_o) \\ &\leq T_{clk} = 0.69 * (14R_oC_o)\ ✓ \end{aligned}$$